

DATASCI 530 – Projects 1 and 2

For these projects, you will be asked to create and analyze a dataset via web scraping. Pick an online marketplace where you can search for products. Your goal is to design a web scraping algorithm that does the following:

Project 1 Deliverables:

- Choose a website that permits automated access and make sure that your scraping process complies with the website's terms and access restrictions.
- Conduct an automated search for a particular type of product. You may choose the type of product you want to analyze.
- Extract information about products (name, vendor, price, rating, detailed product description, and product categories). You do not need to extract all these characteristics depending on your website. The data you extract should be stored in one or more datasets.
- Document your web scraping process, including when the data were collected, how many observations you collected, and any restrictions you placed on the data collection process.
- Your algorithm should include automated navigation: There should be at least one instance in which your algorithm clicks a button or link, navigates to another page, and extracts information from that page. Navigation should be automated via loops or parallelization functions.

Project 2 Deliverables:

- Address any feedback on your Project 1 deliverable.
- Using the data collected in Project 1, produce key descriptive statistics and interpret them.
- In your report, include an analysis of what we learn from this data.

You will have to submit an HTML file with your results and a CSV file with the data that you extracted.

Note: You can also choose to scrape a website that is not a marketplace. You can discuss with the instructor whether your choice of website meets the project requirements and agree on the information to extract ahead of time.

Rubric for Project 1:

You will be graded both on the code and the writing using the following point system.

The use of generative AI is encouraged for diagnosing problems, debugging code, and learning how to use new packages. However, AI use should be focused on specific tasks, and AI-generated code should be reviewed and edited to suit the needs of the project. The final product should be polished, concise, and readable. Excessively long or unedited AI-generated code may receive fewer points.

Component	Detailed Points	Total Points
Project Description Overview: What is the source of the data, when the data was collected, what data within the website was collected, and decisions about how to restrict the data collection process.	3	3
Scraping Algorithm (a) Quality of code (no errors, proper use of web scraping tools). (b) Automated data collection using loops. (c) Data are extracted from multiple items (for example, multiple products in a marketplace). (d) Algorithm includes automated navigation (there is an instance of clicking on a button or link, going to another page, and extracting information). (e) Raw data is converted to a usable dataset (dataset is included with submission).	2 3 2 2 3	12
Results (a) Quality checks are documented. (b) Present a table reporting the number of observations and missing values. Report the mean and variance of relevant numeric variables.	2 2	4
Discussion Clarity and conciseness.	1	1
Total		20

Rubric for Project 2:

You will be graded on both the code and the writing using the following point system. The second deliverable should primarily focus on (i) addressing any feedback from Project 1 and (ii) producing a polished report.

The report should import the dataset from Project 1 in a separate Jupyter notebook and present final results intended for a non-coder audience.

Component	Detailed Points	Total Points
Overall		3
(a) Organization and documentation of code	2	
(b) Aesthetics	1	
Introduction		2
(a) Project Description	1	
(b) Summarize findings (There should be 3 or 4 key findings summarized at the beginning of the document)	1	
Data Collection Process		4
Address feedback on Project 1 deliverable	4	
Results		10
(a) Tables presenting key information. These should include summary statistics and the number of observations. Tables are properly formatted (e.g., have appropriately rounded values, relabeled columns, captions, and are not too long).	4	
(b) Figures presenting key information about categorical and/or numeric variables. There should be a strong emphasis on interpreting the patterns and insights revealed by the data. All graphs look professional (have axis and title labels, a suitable color scheme, appropriate font sizes, a legend when needed, etc.).	6	
Discussion		
Clarity and conciseness	1	1
Total		20